

Solutions to Mock S(A)MC - Kevin Hu

Question 1

Problem: Value of $\frac{2026^2 - 2025^2}{20.26 + 20.25}$?

Solution. Num: $(2026 - 2025)(2026 + 2025) = 4051$. Denom: 40.51.

Result: $\frac{4051}{40.51} = 100$. **Ans: C**

□

Question 2

Problem: Smallest n s.t. $2026 \mid n!$?

Solution. $2026 = 2 \times 1013$. 1013 is prime, so 1013! is the smallest factorial containing it. 1013! also contains 2. **Ans: B**

□

Question 3

Problem: $a_1 = 1, a_{n+1} = \frac{1}{1+a_n}$. Find a_5 .

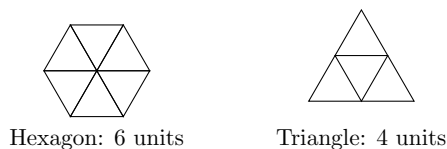
Solution. $a_1 = 1, a_2 = \frac{1}{2}, a_3 = \frac{2}{3}, a_4 = \frac{3}{5}, a_5 = \frac{5}{8}$. **Ans: B**

□

Question 4

Problem: Area ratio of Hexagon to Triangle with equal perimeters ($P = 6$).

Solution. We can construct a regular hexagon with sides $s = 1$ split into 6 identical unit triangles, and an equilateral triangle ($s = 2$) split into 4 unit triangles by connecting midpoints, as shown below.



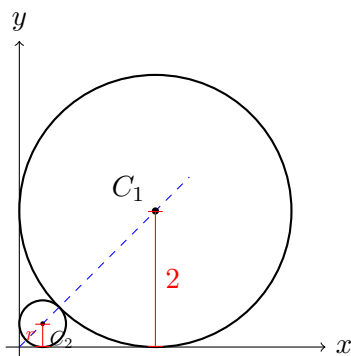
Clearly, the area ratio = $6 : 4 = 3/2$. **Ans: B**

□

Question 5

Problem: Radius r of small circle tangent to large circle ($R = 2$) and axes.

Solution. The center of the large circle is $(2, 2)$ and the center of the small circle is (r, r) .



Origin to large center distance: $2\sqrt{2} = r\sqrt{2} + r + 2$.

$$r(\sqrt{2} + 1) = 2\sqrt{2} - 2 \implies r = 2(\sqrt{2} - 1)^2 = 6 - 4\sqrt{2}. \text{ Ans: A}$$

□

Question 6

Problem: Number of integers n satisfying $n^4 - 20n^2 + 64 < 0$.

Solution. Let $x = n^2$. The inequality $x^2 - 20x + 64 < 0$ factors as $(x - 4)(x - 16) < 0$, giving $4 < x < 16$. Substituting back $x = n^2$, we have $4 < n^2 < 16$. The only integer n whose square is strictly between 4 and 16 is $n = \pm 3$. Thus, there are 2 such integers. **Ans: A** □

Question 7

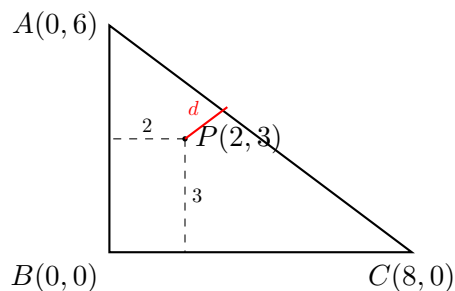
Problem: $N = \overline{abc}$ such that $a + b + c = 15$ and $\overline{cba} - \overline{abc} = 495$.

Solution. Reversal condition: $(100c + 10b + a) - (100a + 10b + c) = 99(c - a) = 495 \implies c - a = 5$. Sum of digits: $a + b + (a + 5) = 15 \implies 2a + b = 10$. Possible (a, b, c) triplets: $(1, 8, 6), (2, 6, 7), (3, 4, 8), (4, 2, 9)$. Note $a = 5 \implies c = 10$ (invalid). Total 4 numbers. **Ans: E** □

Question 8

Problem: $AB = 6, BC = 8, AC = 10$. P is dist 2 from AB , 3 from BC . Find dist to AC .

Solution. Place B at $(0, 0)$, A at $(0, 6)$, and C at $(8, 0)$. Then $P = (2, 3)$. Line AC equation: $\frac{x}{8} + \frac{y}{6} = 1 \implies 3x + 4y - 24 = 0$.



$$\text{Distance } d = \frac{|3(2) + 4(3) - 24|}{\sqrt{3^2 + 4^2}} = \frac{|6 + 12 - 24|}{5} = \frac{6}{5} = 1.2. \text{ Ans: D}$$

□

Question 9

Problem: Sum $S = \sum_{k=1}^{2025} f\left(\frac{k}{2026}\right)$ where $f(x) = \frac{4^x}{4^x + 2}$.

Solution. Note $f(x) + f(1 - x) = \frac{4^x}{4^x + 2} + \frac{4^{1-x}}{4^{1-x} + 2} = \frac{4^x}{4^x + 2} + \frac{2}{2 + 4^x} = 1$. Pairing $f(\frac{k}{2026})$ with $f(\frac{2026-k}{2026})$ gives 1012 pairs of sum 1, plus a middle term $f(1/2) = \frac{2}{2+2} = 0.5$. Total sum = $1012 + 0.5 = 1012.5$.

Ans: B □

Question 10

Problem: Probability the sum of Die A $\{1, 2, 2, 3, 3, 4\}$ and Die B $\{1, 3, 4, 5, 6, 8\}$ is 5.

Solution. Total outcomes: $6 \times 6 = 36$. We construct a grid of all possible sums:

A \ B	1	3	4	5	6	8
1	2	4	5	6	7	9
2	3	5	6	7	8	10
2	3	5	6	7	8	10
3	4	6	7	8	9	11
3	4	6	7	8	9	11
4	5	7	8	9	10	12

The sum of 5 occurs in 4 instances (highlighted).

Probability = $\frac{4}{36} = \frac{1}{9}$. **Ans: B**

□

Question 11

Problem: Sum of real t for acute A, B if $\sin^2 A + \cos^2 B = \frac{5}{4}t$ and $\cos^2 A + \sin^2 B = \frac{3}{4}t^2$.

Solution. Adding the two equations: $(\sin^2 A + \cos^2 A) + (\sin^2 B + \cos^2 B) = \frac{5}{4}t + \frac{3}{4}t^2$.

$$1 + 1 = \frac{3t^2 + 5t}{4} \implies 3t^2 + 5t - 8 = 0.$$

Factoring: $(3t + 8)(t - 1) = 0$, so $t = 1$ or $t = -8/3$.

Since A, B are real, $\sin^2 A + \cos^2 B \geq 0$, so $\frac{5}{4}t \geq 0 \implies t \geq 0$.

Thus, $t = 1$ is the only valid solution. Sum = 1. **Ans: A**

□

Question 12 (Intended for integers not positive integers)

Solution. Rewrite the equation by completing the square:

$$(x + y)^2 + 2y^2 = 34$$

Since $(x + y)^2 \geq 0$, we must have $2y^2 \leq 34$, which implies $y^2 \leq 17$. The possible integer values for y are $\{0, \pm 1, \pm 2, \pm 3, \pm 4\}$. We test each case for $(x + y)^2 = 34 - 2y^2$:

1. $y = 0 \implies (x + 0)^2 = 34$ (No integer solution)
2. $y = \pm 1 \implies (x \pm 1)^2 = 32$ (No integer solution)
3. $y = \pm 2 \implies (x \pm 2)^2 = 26$ (No integer solution)
4. $y = \pm 3 \implies (x \pm 3)^2 = 34 - 18 = 16 \implies x \pm 3 = \pm 4$
 - If $y = 3$: $x + 3 = 4 \implies x = 1$; or $x + 3 = -4 \implies x = -7$.
 - If $y = -3$: $x - 3 = 4 \implies x = 7$; or $x - 3 = -4 \implies x = -1$.
5. $y = \pm 4 \implies (x \pm 4)^2 = 34 - 32 = 2 \implies$ (No integer solution)

The complete set of integer solutions (x, y) is:

$$\{(1, 3), (-7, 3), (7, -3), (-1, -3)\}$$

There are 4 such pairs. **Ans: D**

□

Question 13

Problem: Min value of $a^4 + ab + b^4$ given $a^2 + b^2 = 1$.

Solution. $a^4 + b^4 = (a^2 + b^2)^2 - 2a^2b^2 = 1 - 2(ab)^2$.

Let $k = ab$. Since $|ab| \leq \frac{a^2+b^2}{2}$, we have $-\frac{1}{2} \leq k \leq \frac{1}{2}$.

Expression $f(k) = 1 - 2k^2 + k$. This is a downward parabola with vertex at $k = \frac{-1}{2(-2)} = 1/4$.

Max occurs at $k = 1/4$, but we want the **minimum**. Check endpoints:

$$f(1/2) = 1 - 2(1/4) + 1/2 = 1.$$

$$f(-1/2) = 1 - 2(1/4) - 1/2 = 0.$$

The minimum value is 0. **Ans: A**

□

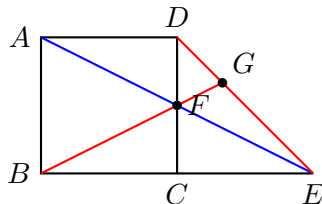
Question 14

Problem: $ABCD$ unit square, $CE = 1$. $AE \cap DC = F$, $BF \cap DE = G$. Find BG .

Solution. We can approach using simple coordinate geometry bash. Place B at $(0,0)$, A at $(0,1)$, C at $(1,0)$, and D at $(1,1)$. Since $CE = 1$ and E is on BC extended, $E = (2,0)$.

1. Find F : Line AE passes through $(0,1)$ and $(2,0) \Rightarrow y = -\frac{1}{2}x + 1$. Since F is on CD ($x = 1$), $y_F = -\frac{1}{2}(1) + 1 = 0.5$. So $F = (1, 0.5)$.

2. Find G : Line BF passes through $(0,0)$ and $(1,0.5) \Rightarrow y = 0.5x$. Line DE passes through $(1,1)$ and $(2,0) \Rightarrow y = -x + 2$. Equating them: $0.5x = -x + 2 \Rightarrow 1.5x = 2 \Rightarrow x = 4/3, y = 2/3$. Thus $G = (4/3, 2/3)$.



3. Calculate BG : $BG = \sqrt{(4/3)^2 + (2/3)^2} = \sqrt{16/9 + 4/9} = \sqrt{20/9} = \frac{2\sqrt{5}}{3}$. **Ans: D**

□

Question 15

Problem: Min value of $f(x) = \sum_{k=0}^{2017} |x - k|$.

Solution. The sum of absolute differences is minimized at the median of the data points.

Points are $\{0, 1, \dots, 2017\}$ (2018 points). Median is any value in $[1008, 1009]$.

Let $x = 1008$. $f(1008) = \sum_{k=0}^{2017} |1008 - k| = 2 \sum_{j=1}^{1008} j + 1009 = 2 \frac{1008 \times 1009}{2} + 1009$.

$1008 \times 1009 + 1009 = 1009^2 = 1018081$. **Ans: B**

□

Question 16

Problem Statement: Let $S = \sum_{n=1}^{2017} \frac{1}{n^3}$. Find the value of $\lfloor 4S \rfloor$.

Solution. To determine $\lfloor 4S \rfloor$, we approximate S by calculating the initial terms and bounding the remaining "tail" of the series.

1. Grouping

We separate the first three terms and group the remaining terms to create an upper bound:

$$S = 1 + \frac{1}{8} + \frac{1}{27} + \sum_{n=4}^{2017} \frac{1}{n^3}$$

$$S < 1 + \underbrace{\left(\frac{1}{2^3} + \frac{1}{3^3}\right)}_{\text{Group 1}} + \underbrace{\left(\frac{1}{4^3} + \cdots + \frac{1}{7^3}\right)}_{\text{Group 2}} + \underbrace{\left(\frac{1}{8^3} + \cdots + \frac{1}{15^3}\right)}_{\text{Group 3}} + \cdots$$

2. Hidden geometric series

By comparing the sum to a geometric series where each block of 2^k terms is bounded by the first term of the block, we get:

$$S < 1 + \frac{1}{8} + \frac{1}{27} + \left[\frac{1}{16} + \frac{1}{64} + \frac{1}{256} + \cdots \right]$$

$$S < 1.1620 + \sum_{k=2}^{\infty} \left(\frac{1}{4}\right)^k$$

The sum of the infinite geometric series $\sum_{k=2}^{\infty} \left(\frac{1}{4}\right)^k$ with first term $a = \frac{1}{16}$ and ratio $r = \frac{1}{4}$ is:

$$\text{Tail} = \frac{1/16}{1 - 1/4} = \frac{1/16}{3/4} = \frac{1}{12} \approx 0.0833$$

3. Approximation

Adding the components together:

$$S < 1.1620 + 0.0833 = 1.2453$$

Multiplying by 4 to find the bounds for $4S$:

- **Upper Bound:** $4S < 4 \times 1.2453 = 4.9812$
- **Lower Bound:** Since $S > 1 + \frac{1}{8} + \frac{1}{27} \approx 1.1620$, then $4S > 4 \times 1.1620 = 4.648$

Since $4.648 < 4S < 4.9812$, the integer part must be 4. One can also add up $1 + \frac{1}{8} + \frac{1}{27} + \frac{1}{12}$, and then multiply by 4, which is fairly trivial to compute.

Ans: B □

Question 17

Problem: Find the mean of all values of $x + \frac{1}{x}$ for $4x^4 + 3x^3 - 2x^2 + 3x + 4 = 0$.

Solution. This is a reciprocal equation. Divide by x^2 :

$$4x^2 + 3x - 2 + \frac{3}{x} + \frac{4}{x^2} = 0 \implies 4\left(x^2 + \frac{1}{x^2}\right) + 3\left(x + \frac{1}{x}\right) - 2 = 0$$

Let $u = x + \frac{1}{x}$. Then $x^2 + \frac{1}{x^2} = u^2 - 2$. Substitute:

$$4(u^2 - 2) + 3u - 2 = 0 \implies 4u^2 + 3u - 10 = 0$$

This is a quadratic in u . The roots u_1, u_2 represent the possible values of $x + \frac{1}{x}$. By Vieta's formulas, the sum of the roots is $u_1 + u_2 = -b/a = -3/4$. The mean is $\frac{u_1 + u_2}{2} = \frac{-3/4}{2} = -3/8$. **Ans: B** □

Question 18

Problem: If $x + \frac{1}{x} = \sqrt{5}$, what is $x^{11} + \frac{1}{x^{11}}$?

Solution. Let $V_n = x^n + x^{-n}$. We use the recurrence $V_n = V_1 V_{n-1} - V_{n-2}$.

$$\begin{aligned} V_0 &= 2 \rightarrow V_1 = \sqrt{5} \rightarrow V_2 = 3 \rightarrow V_3 = 2\sqrt{5} \\ \dots V_n &= \sqrt{5}V_{n-1} - V_{n-2} \dots \end{aligned}$$

Iterating: $V_4 = 7$, $V_5 = 5\sqrt{5}$, $V_6 = 18$. Using the identity $V_{m+n} = V_m V_n - V_{m-n}$:

$$V_{11} = V_6 V_5 - V_1 = (18)(5\sqrt{5}) - \sqrt{5} = 90\sqrt{5} - \sqrt{5} = 89\sqrt{5}$$

Ans: C

□

Question 19

Problem: $P(k) = \frac{k}{k+1}$ for $k = 0, \dots, n$. Find $P(n+1)$.

Solution. Define $Q(x) = (x+1)P(x) - x$. $Q(x)$ is a polynomial of degree $n+1$ with roots at $x = 0, 1, \dots, n$. Thus, $Q(x) = C \cdot x(x-1)(x-2) \dots (x-n)$. To find C , let $x = -1$:

$$Q(-1) = 0 - (-1) = 1 \implies 1 = C(-1)(-2) \dots (-(n+1)) = C(-1)^{n+1}(n+1)!$$

So $C = \frac{(-1)^{n+1}}{(n+1)!}$. Now evaluate $Q(n+1)$:

$$(n+2)P(n+1) - (n+1) = \frac{(-1)^{n+1}}{(n+1)!}(n+1)! = (-1)^{n+1}$$

Solving for $P(n+1)$:

$$P(n+1) = \frac{n+1 + (-1)^{n+1}}{n+2}$$

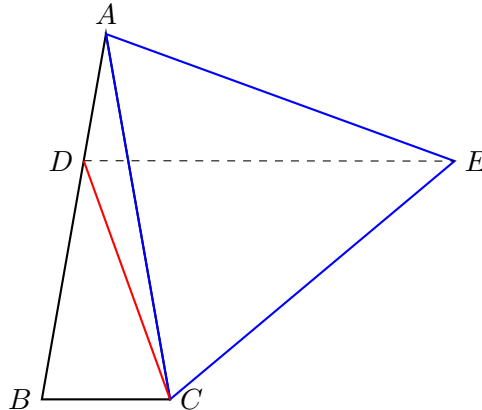
Ans: C

□

Question 20

Problem: In isosceles $\triangle ABC$, $\angle A = 20^\circ$ and $AB = AC$. D lies on AB such that $AD = BC$. Find $\angle BDC$.

Solution 1 - pure geometry. In $\triangle ABC$, $\angle B = \angle C = 80^\circ$. Construct equilateral $\triangle ACE$ externally. Connect DE and DC .



1. In $\triangle ADE$ and $\triangle CAB$: $AD = BC$ (given), $AE = AC$ (equilateral), and $\angle EAD = 60^\circ + 20^\circ = 80^\circ$. By SAS, $\triangle ADE \cong \triangle CAB$.
2. From congruence, $ED = AB$. Since $AB = AC = EC$, $\triangle DEC$ is isosceles with $ED = EC$.
3.
 - $\angle ADE = \angle ABC = 80^\circ$ and $\angle AED = \angle BAC = 20^\circ$.
 - $\angle DEC = \angle AEC - \angle AED = 60^\circ - 20^\circ = 40^\circ$.
 - Thus, the base angle $\angle EDC = (180^\circ - 40^\circ)/2 = 70^\circ$.
4. $\angle BDC = 180^\circ - \angle ADE - \angle EDC = 180^\circ - 80^\circ - 70^\circ = 30^\circ$.

Ans: C

□

Solution 2 - Trig bash (Not Recommended). Normalize by letting $BC = AD = 1$. In $\triangle ABC$, the Sine Rule gives:

$$\frac{1}{\sin 20^\circ} = \frac{AB}{\sin 80^\circ} \implies AB = \frac{\sin 80^\circ}{\sin 20^\circ}$$

In $\triangle BDC$, we have $\angle B = 80^\circ$ and $BD = AB - 1$. By the Sine Rule:

$$\frac{BD}{\sin(100^\circ - x)} = \frac{1}{\sin x} \implies \sin x(AB - 1) = \sin(100^\circ - x)$$

Substitute AB :

$$\sin x \left(\frac{\sin 80^\circ - \sin 20^\circ}{\sin 20^\circ} \right) = \sin(100^\circ - x)$$

Using $\sin 80^\circ - \sin 20^\circ = 2 \cos 50^\circ \sin 30^\circ = \cos 50^\circ = \sin 40^\circ$:

$$\sin x \left(\frac{\sin 40^\circ}{\sin 20^\circ} \right) = \sin x(2 \cos 20^\circ) = \sin(100^\circ - x)$$

Expanding the right side:

$$2 \sin x \cos 20^\circ = \sin 100^\circ \cos x - \cos 100^\circ \sin x$$

Divide by $\sin x$ and use $\sin 100^\circ = \cos 10^\circ$, $\cos 100^\circ = -\sin 10^\circ$:

$$2 \cos 20^\circ = \cos 10^\circ \cot x + \sin 10^\circ$$

$$\cot x = \frac{2 \cos 20^\circ - \sin 10^\circ}{\cos 10^\circ} = \frac{\sqrt{3} \cos 10^\circ + \sin 10^\circ - \sin 10^\circ}{\cos 10^\circ} = \sqrt{3}$$

Thus, $x = 30^\circ$. **Ans: C**

□

Question 21

Problem: If a, b, c are roots of $x^3 - 5x^2 + 7x - 9 = 0$, find $S = \frac{a^2}{bc} + \frac{b^2}{ac} + \frac{c^2}{ab}$.

Solution. The expression simplifies to $S = \frac{a^3 + b^3 + c^3}{abc}$. From Vieta's formulas:

- $e_1 = a + b + c = 5$
- $e_2 = ab + bc + ca = 7$
- $e_3 = abc = 9$

Using the identity $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - (ab + bc + ca))$:

1. $a^2 + b^2 + c^2 = e_1^2 - 2e_2 = 5^2 - 2(7) = 11$.
2. $a^3 + b^3 + c^3 - 3(9) = 5(11 - 7) \implies a^3 + b^3 + c^3 - 27 = 20$.
3. $a^3 + b^3 + c^3 = 47$.

Thus, $S = \frac{47}{9}$. **Ans: B** □

Question 22

Problem: The number $N = \underbrace{11 \dots 1}_{81}$ is divisible by which power of 3?

Solution. A repunit R_n is divisible by 3^k if n is divisible by 3^{k-1} (for $n > 1$). Formally, using the Lifting The Exponent Lemma for $p = 3$:

$$v_3(R_n) = v_3\left(\frac{10^n - 1}{9}\right) = v_3(10^n - 1) - v_3(9)$$

Since $v_3(10 - 1) = 2$, for $n = 81$:

$$v_3(10^{81} - 1) = v_3(10 - 1) + v_3(81) = 2 + 4 = 6$$

Then $v_3(N) = 6 - 2 = 4$. So N is divisible by $3^4 = 81$. **Ans: A** □

Question 23

Problem: Sum of p for $x^2 + 2px - 3p - 2 = 0$ if $(x_1^3 + x_2^3) + (x_1^2 + x_2^2) = 4$.

Solution. Vieta's: $s = x_1 + x_2 = -2p$ and $q = x_1x_2 = -3p - 2$.

- $x_1^2 + x_2^2 = s^2 - 2q = 4p^2 + 6p + 4$.
- $x_1^3 + x_2^3 = s(s^2 - 3q) = -2p(4p^2 + 9p + 6) = -8p^3 - 18p^2 - 12p$.

Equation: $(-8p^3 - 18p^2 - 12p) + (4p^2 + 6p + 4) = 4 \implies -8p^3 - 14p^2 - 6p = 0$. Factoring: $-2p(4p+3)(p+1) = 0$. The values are $p = 0, p = -1, p = -3/4$. The discriminant $\Delta = 4(p+1)(p+2)$ must be > 0 for real distinct roots. This excludes $p = -1$. Remaining $p \in \{0, -3/4\}$. Sum = $-3/4$.

Ans: B □

Question 24

Problem: $abc = -1, a + b + c = 4, \sum \frac{a}{a^2 - 3a - 1} = 4/9$. Find $a^2 + b^2 + c^2$.

Solution. Since $abc = -1$, the denominator $a^2 - 3a - 1$ can be written as $a^2 - 3a + abc = a(a - 3 + bc)$. As $a - 3 = 1 - b - c$, the expression becomes $a(bc - b - c + 1) = a(b - 1)(c - 1)$. The sum is:

$$\sum_{cyc} \frac{1}{(b-1)(c-1)} = \frac{(a-1) + (b-1) + (c-1)}{(a-1)(b-1)(c-1)} = \frac{(a+b+c) - 3}{abc - (ab + bc + ca) + (a+b+c) - 1}$$

Let $x = ab + bc + ca$. Then:

$$\frac{4 - 3}{-1 - x + 4 - 1} = \frac{1}{2 - x} = \frac{4}{9} \implies x = -1/4$$

$a^2 + b^2 + c^2 = x^2 - 2x = 16 - 2(-1/4) = \frac{33}{2}$. **Ans: E** □

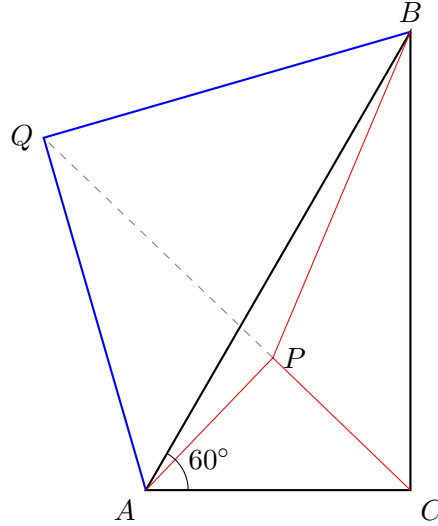
Question 25

Problem Statement: In $\triangle ABC$, $\angle A = 60^\circ$ and $AB = 2AC$. Let P be a point **inside** the triangle such that $PA = \sqrt{3}$, $PB = 5$, and $PC = 2$. Find the area of $\triangle ABC$.

Solution. Since $AB = 2AC$ and $\angle A = 60^\circ$, $\triangle ABC$ is a 30° - 60° - 90° triangle with $\angle ACB = 90^\circ$ and $\angle ABC = 30^\circ$.

1. Construction

Construct $\triangle ABQ$ outside $\triangle ABC$ such that $\triangle ABQ \sim \triangle ACP$ via a spiral similarity centered at A . Specifically, $\angle QAB = \angle PAC$ and $\frac{AQ}{AB} = \frac{AP}{AC}$. Since $AB/AC = 2$, it follows that $AQ = 2AP = 2\sqrt{3}$ and $BQ = 2CP = 4$.



2. Intermediate Relationships

The angle $\angle QAP = \angle QAB + \angle BAP = \angle PAC + \angle BAP = \angle BAC = 60^\circ$.

In $\triangle APQ$, since $AQ = 2AP$ and $\angle QAP = 60^\circ$, it is a 30° - 60° - 90° triangle. Thus:

- $PQ = \sqrt{3}AP = \sqrt{3}(\sqrt{3}) = 3$.
- $\angle APQ = 90^\circ$.

3. Cosine Rule

In $\triangle BQP$, the sides are $BQ = 4$, $PQ = 3$, and $PB = 5$. Since $3^2 + 4^2 = 5^2$, $\angle BQP = 90^\circ$ (DIAGRAM NOT TO SCALE) by the Pythagorean converse.

Now, in $\triangle ABQ$, $\angle AQB = \angle AQP + \angle BQP = 30^\circ + 90^\circ = 120^\circ$.

By the Law of Cosines in $\triangle ABQ$:

$$AB^2 = AQ^2 + BQ^2 - 2(AQ)(BQ)\cos(120^\circ)$$

$$AB^2 = (2\sqrt{3})^2 + 4^2 - 2(2\sqrt{3})(4)(-1/2)$$

$$AB^2 = 12 + 16 + 8\sqrt{3} = 28 + 8\sqrt{3}$$

4. Area

For a 30° - 60° - 90° triangle with hypotenuse AB :

$$\text{Area} = \frac{1}{2} \cdot \left(\frac{AB}{2}\right) \cdot \left(\frac{AB\sqrt{3}}{2}\right) = \frac{\sqrt{3}}{8}AB^2$$

$$\text{Area} = \frac{\sqrt{3}}{8}(28 + 8\sqrt{3}) = \frac{28\sqrt{3} + 24}{8} = \frac{7\sqrt{3} + 6}{2}$$

Ans: E

□